

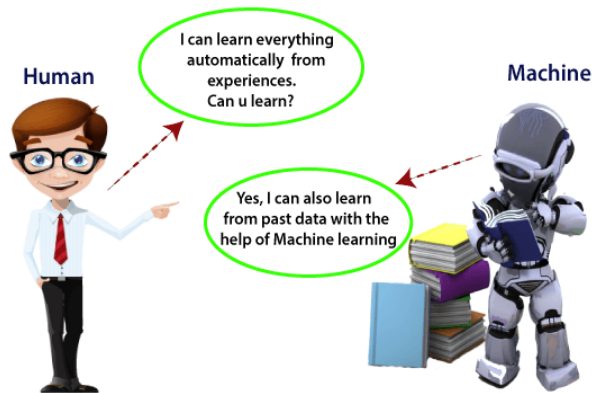
INTRODUCTION TO MACHINE LEARNING

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What is machine learning :

- In the real world, we are surrounded by humans who can learn everything from their experiences with their learning capability
- We have computers (or) machines which work on our instructions
- But can a machine also learn from experiences (or) past data like a human does?
- So here comes the role of machine learning

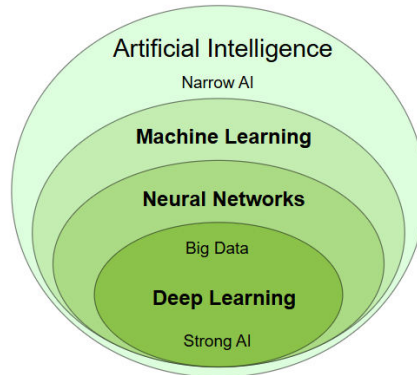


Ex :

- Imagine you are teaching a child to recognize different animals
- Now you show the child pictures of different animals like a dog, cat, rabbit and tell them the names of each one
- After seeing many pictures, the child starts recognizing patterns like dogs have certain shapes, cats have different ears and so on
- One day, the child sees a new animal (a dog they have never seen before) and can still say “ That’s a dog! ” because they learned the features of a dog from the pictures you showed them earlier
- In machine learning, instead of a child, you have a computer
- You give the computer many examples like pictures of animals and the machine learns to recognize patterns in the data
- After enough learning, the machine can recognize new things, just like the child recognized the dog

Introduction to machine learning :

- Machine learning is a subfield of artificial intelligence



- Machine primarily focuses on the creation of algorithms that enable a computer to independently learn from data and previous experiences
- Arthur Samuel first used the term “ machine learning ” in 1959
- Without being explicitly programmed, machine learning enables a machine to automatically learn from data and improve performances from experiences and predict things
- Machine learning algorithms create a mathematical model that, without being explicitly programmed, aids in making predictions (or) decisions with the assistance of sample historical data (or) training data

Note :

- Traditional programming uses algorithms to produce results from data

Data + Algorithms = Results

- Machine learning creates algorithms from data and results

Data + Results = Algorithms

Learning problems :

Problem :

- A task which need to be solved
(or)
- A problem is a situation where some something needs to be solved (or) improved
- It is a challenge that requires a solution

Ex :

- Imagine you are hungry but have no food at home
 - Problem – You need to eat, but there is no food
 - Possible solutions – Cook something, order food, go to a restaurant and so on

- A problem exists when there is a gap between the current situation and the desired outcomes
- Solving the problem means finding a way to close that gap

Learning problem :

- The problems that can effectively solved by using machine learning
(or)
- Learning problem is a situation where a system (or) a person needs to improve its performance on a task based on experience

Well-posed learning problem :

- A learning problem which have three features
 - Task (T) – Objective
 - Performance measure (P) – Metric used to assess the effectiveness of the algorithm at the task
 - Training Experience (E) – Dataset from which algorithm can learn

Ex :

- Imagine we are building a self-driving robot that learns to drive on its own
 - Task (T) – The robot needs to drive safely on the road without crashing
 - Performance measure (P) – The success is measured by how well it avoids obstacles, follows traffic rules and reaches the destination
 - Experience (E) – The robot learns from data (camera images, sensor readings) and past driving experiences (successes and mistakes)
- Overtime, the robot improves its driving skills by learning from its experiences just like a human driver

Note :

- Learning scenarios – Supervised learning (Classification and regression), Unsupervised learning (Clustering and dimensionality reduction), Semi-supervised learning (Image labeling), Reinforcement learning (Game playing and robotics)

Need for machine learning :

- Machine learning is important because it allows computers to learn from data and improve their performance on specific tasks without being explicitly programmed
- This ability to learn from data and adapt to new situations makes machine learning particularly useful for tasks that involve large amounts of data, complex decision-making and so on
- Here are some specific areas where machine learning is being used
 - Predictive modeling

- Natural language processing
- Computer vision
- Fraud detection
- Recommendation systems

Predictive modeling :

- Predictive modeling is a process used in data science to create a mathematical model that predicts an outcome based on input data
- It involves using statistical algorithms and machine learning techniques to analyze historical data make predictions about future (or) unknown events

Natural language processing :

- Natural language processing is a field of computer science and a subfield of artificial intelligence that aims to make computers understand human language
- NLP uses computational linguistics, which is the study of how language works and various models based on statistics, machine learning and deep learning
- These techniques allow computers to analyze and process text (or) voice data and grasp their full meaning, including the speaker's (or) writer's intentions and emotions

Computer vision :

- Computer vision is a way for computers to “ see ” and understand pictures (or) videos just like humans do
- Machine learning helps computers learn from thousands of images, so they can recognize objects, faces, read text from pictures and so on
- This is important for applications such as self-driving cars, surveillance systems and medical imaging

Fraud detection :

- Fraud detection using machine learning helps find suspicious (or) dishonest activities by looking for unusual patterns in data
- The system learns from past fraud cases and then watches for similar patterns in new transactions
- Machine learning can be used to detect fraudulent behavior in financial transactions, online advertising and other areas

Recommendation systems :

- Recommendation system is like a smart assistant that suggests things you might like based on what you have watched, bought and searched for before

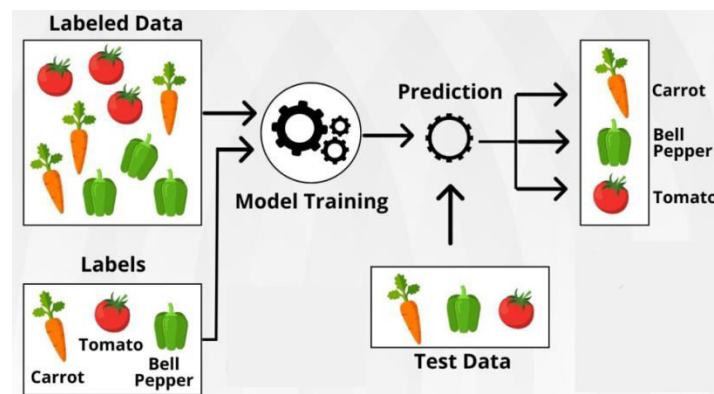
- Machine learning helps it learn from your past choices and the choices of others with similar interests
- For example, online shopping applications like Amazon, Flip kart and so on .

Types of learning :

- Machine learning is the branch of artificial intelligence that focuses on developing models and algorithms that let computers learn from data and improve from previous experience without being explicitly programmed for every tasks
- In simple words, machine learning teaches the systems to think and understand like humans by learning from the data
- There are several types of machine learning, each with special characteristics and applications
- Some of the types of machine learning algorithms are as follows
 - Supervised machine learning
 - Unsupervised machine learning
 - Semi supervised learning
 - Reinforcement learning

Supervised machine learning :

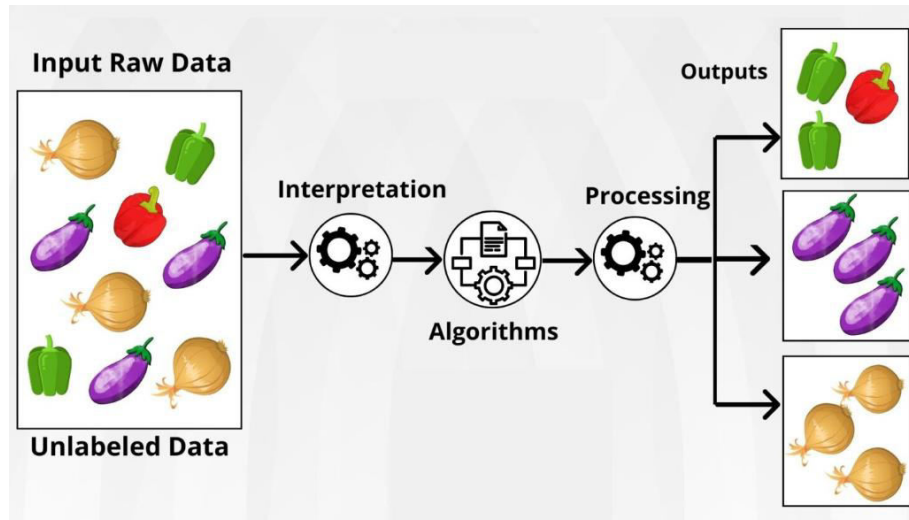
- Supervised learning is based on supervision
- It means in the supervised learning technique, we train the machines using “ labeled “ dataset and based on the training, the machine predicts the output
- Here, the labeled data specifies that some of the inputs are already mapped to the output
- More preciously, we can say, we train the machine with the input and corresponding output and then we ask the machine to predict the output using the test dataset



Unsupervised machine learning :

- Unsupervised learning is different from the supervised learning techniques, as its name suggests, there is no supervision
- It means, in unsupervised machine learning, the machine is trained using the unlabeled dataset and the machine predicts the output without any supervision

- In unsupervised learning, the models are trained with the data that is neither nor labeled and the model acts on that data without any supervision
- The main aim of the unsupervised learning algorithm is to group (or) categories the unsorted dataset according to the similarities, patterns and differences
- Machines are instructed to find the hidden patterns from the input dataset



Note :

- Interpretation – The system looks at the input data and tries to understand it and it does not know what each item is, but it tries to find patterns
- Algorithms (Brain of the system) – Algorithm is the set of rules that help the system analyze the data and it looks for similarities between objects and decides how to group them
- Processing – The system applies the algorithm to the data and organizes it into groups and similar items are placed together based on their shared features

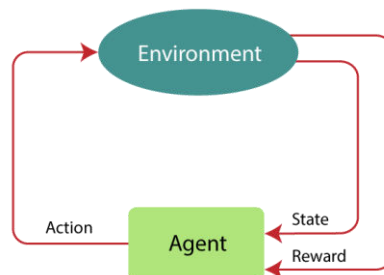
Semi supervised learning :

- Semi-supervised learning is a type of machine learning algorithm that lies between supervised and unsupervised machine learning
- It represents the intermediate ground between supervised (with labeled training data) and unsupervised learning (with no labeled training data) algorithms and uses the combination of labeled and unlabeled datasets during the training period
- Although semi-supervised learning is the middle ground between supervised and unsupervised learning and operates on the data that consists of a few labels, it mostly consists of unlabeled data
- As labels are costly, but for corporate purposes, they may have few labels
- It is completely different from supervised and unsupervised learning as they are based on the presence and absence of labels

- To overcome the drawbacks of supervised learning and unsupervised learning algorithms, the concept of semi-supervised learning is introduced
- The main aim of semi-supervised learning is to effectively use all the available data, rather than only labeled data like in supervised learning

Reinforcement learning :

- Reinforcement learning is a feedback based machine learning technique in which an agent learns to behave in an environment by performing the actions and seeing the results of actions
- For each good action, the agent gets positive feedback and for each bad action, the agent gets negative feedback (or) penalty
- There is no labeled data, so the agent is bound to learn by its experience only
- Reinforcement learning solves a specific type of problem where decision making is sequential and the goal is long-term such as game playing, robotics and so on



Ex :

- Imagine you are training a dog to sit
 - Environment – Your home
 - Agent – The dog
 - State – Whether the dog is standing, sitting, lying down
 - Action – The dog decides to sit (or) not
 - Reward – If the dog sits when you told, you give it a treat (positive reward) and if not, no treat (negative feedback)
- Over time, the dog learns that sitting when you say "sit" results in a treat, so it starts sitting more often
- Similarly in reinforcement learning, an AI model learns by repeating actions and receiving feedback, improving over time

Note :

- Reinforcement learning is a machine learning training method based on a reward and punishment mechanism
- Agent is rewarded for correct moves and punished for the wrong ones
- In doing so, the agent tries to minimize wrong moves and maximize the right ones

Standard learning tasks :

- Classification
- Regression
- Clustering
- Dimensionality reduction
- Outlier detection
- Association rule learning
- Reinforcement learning

ASSOCIATION RULES

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Association analysis :

- Association analysis is a technique used to find patterns (or) relationships in large sets of data, especially in transactions
- The goal is to discover items that frequently occur together
- These relationships can be written as association rules, while help to understand the connection between different items (or) events
- For example, let's say you have a grocery store and you want to know which items are often bought together and You collect data from customers shopping baskets

Ex :

- Imagine the following transactions at the store
 - Transaction 1 : Milk, Bread
 - Transaction 2 : Milk, Butter
 - Transaction 3 : Bread, Butter
 - Transaction 4 : Milk, Bread, Butter
- Frequent item sets – First, we find items that appear together in many transactions, In our example
 - Milk and bread are bought together in 2 out of 4 transactions
 - Milk and Butter are bought together in 2 out of 4 transactions
 - Bread and Butter are bought together in 2 out of 4 transactions
- Association rules – From these frequent item sets, we create rules like
 - “ If milk, then bread ”
 - “ If milk, then butter ”
 - “ If bread, then butter ”
- By finding these patterns, a store can make decisions on how to promote products together (or) where to place them for more sales

Frequent pattern mining :

- Frequent pattern mining is a technique used to find patterns that occur often in a set of data
- These patterns could be items, actions, or events that happen together frequently
- It's like looking for regularities (or) trends that show up over and over

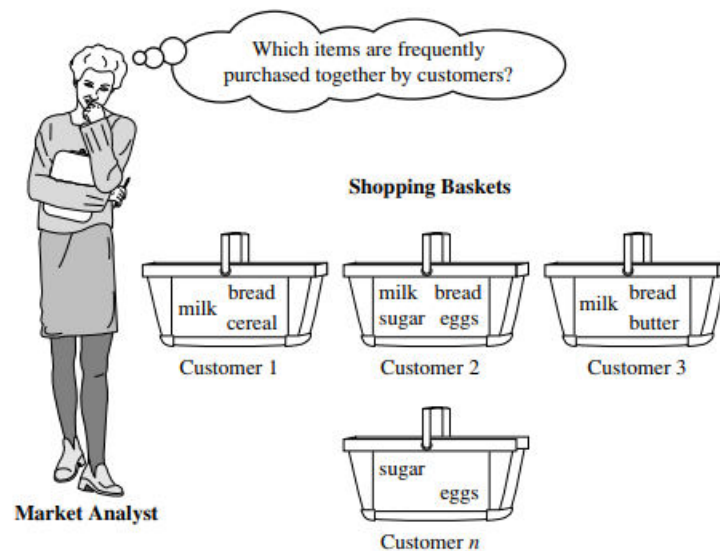
Ex :

- Let's say we have a set of shopping transactions from a store
 - Transaction 1: Milk, Bread
 - Transaction 2: Milk, Butter
 - Transaction 3: Bread, Butter
 - Transaction 4: Milk, Bread, Butter
- In frequent pattern mining, we look for items that appear together in many transactions

- Count the occurrences
 - Milk and Bread appear together in 2 out of 4 transactions
 - Milk and Butter appear together in 2 out of 4 transactions
 - Bread and Butter appear together in 2 out of 4 transactions
 - Milk, Bread, and Butter appear together in 1 out of 4 transactions
- Find frequent patterns
 - Milk and Bread are a frequent pattern because they appear together in at least 2 transactions
 - Milk and Butter are also a frequent pattern

Market basket analysis :

- Market Basket Analysis is a technique used in data mining to understand the purchasing behavior of customers by analyzing the items they buy together
- It's like studying the contents of a customer's shopping cart to find patterns and relationships between products



Association rule :

- Association rule describes the relationship between 2 items in a dataset
- It's often used in market basket analysis to find patterns
- If there is a pair of items are X and Y, which are frequently bought together then association rule is represented as $X \Rightarrow Y$

Ex :

- Customers who purchase computers also tend to buy antivirus software at the same time is represented as

Computer $\Rightarrow$ Antivirus_software

Measures of association rules :

- There are two main measures used to evaluate the strength of an association rule
 - Support
 - Confidence

Support :

- Support tells us how frequently an item set (or) combination of items appears in the dataset
- Support can be represented by using the formula

$$\text{Support} = \frac{\text{Number of transactions containing the item set}}{\text{Total number of transactions}}$$

(or)

$$\text{Support}\{A\} = \frac{\text{Frequency}\{A\}}{N}$$

(or)

$$\text{Support}\{A,B\} = \frac{\text{Frequency}\{A,B\}}{N}$$

- Where A, B are items and N is the total number of transactions

Ex :

- Customer transactions are shown in the below table

TID	Items
T1	Bread, Coke, Milk
T2	Popcorn, Bread
T3	Bread, Egg, Milk
T4	Egg, Bread, Coke, Milk
T5	Egg, Apple

- Support for item coke

$$\text{Support}\{\text{Coke}\} = \frac{\text{Frequency}\{\text{Coke}\}}{N}$$

$$= 2 / 5$$

$$= 0.4 \text{ (or) } 40\%$$

Confidence :

- This says that how likely item B is purchased when item A is purchased
- It can be expressed as $\{ A \rightarrow B \}$
- Confidence of items (A and B) is the frequency (or) number of transactions in which the items (A and B) appear, divided by the frequency (or) number of transactions in which the item (A) appears
- Confidence can be represented by using the formula

$$\text{Confidence} \{ A \rightarrow B \} = \frac{\text{Frequency} \{ A, B \}}{\text{Frequency} \{ A \}}$$

(or)

$$\text{Confidence} \{ A \rightarrow B \} = \frac{\text{Support} \{ A, B \}}{\text{Support} \{ A \}}$$

Ex :

- Customer transactions is shown in the below table

TID	Items
T1	Bread, Coke, Milk
T2	Popcorn, Bread
T3	Bread, Egg, Milk
T4	Egg, Bread, Coke, Milk
T5	Egg, Apple

- Confidence of $\{ \text{Bread} \rightarrow \text{Milk} \}$ is

$$\text{Confidence} \{ \text{Bread} \rightarrow \text{Milk} \} = \frac{\text{Frequency} \{ \text{Bread, Milk} \}}{\text{Frequency} \{ \text{Bread} \}}$$

$$= 3 / 4$$

$$= 0.75$$

$$= 75\%$$

(or)

- Confidence of $\{ \text{Bread} \rightarrow \text{Milk} \}$ is

$$\text{Support of} \{ \text{Bread, Milk} \} = 3/5$$

$$= 0.6$$

$$\text{Support of} \{ \text{Bread} \} = 4/5$$

$$= 0.8$$

$$\begin{aligned}\text{Confidence \{ Bread } \rightarrow \text{ Milk \}} &= \frac{\text{Support \{ Bread, Milk \}}}{\text{Support \{ Bread \}}} \\ &= 0.6 / 0.8 \\ &= 0.75 \\ &= 75\%\end{aligned}$$

Applications of market basket analysis :

Retail :

- Market basket analysis helps stores understand what products are bought together
- For example, if people often buy soda and chips together, the store might place them near each other to encourage more sales
- On the other hand, if people don't usually buy soda and crackers together, it may not make sense to place them together on the same shelf
- So, it's about grouping products that people tend to buy at the same time

Telecommunications :

- In this industry, companies can use market basket analysis to see which services people buy together
- For example, many customers might buy TV and Internet together, so the company might create special promotions for that combinations to attract more customers

Banks :

- Banks use market basket analysis to look at the purchases customers make with their credit cards
- For example, if you always buy groceries and pay for gas with your credit card, the bank can notice that patterns
- If suddenly, there is a big purchase from an unfamiliar location, it might trigger an alert for possible fraud

Insurance :

- Insurance companies use market basket analysis to detect fraudulent claims
- For example, if a person files multiple medical claims in one month for injuries that are too similar, the insurance company might investigate to see if it is a fake claim [i.e., Insurance company will check if it's a fake claim (or) not]

Frequent item set generation :

- Frequent item set mining is a method for market basket analysis
- Frequent item set mining is the process of finding combinations of items that are often bought together in transactions
- It helps businesses understand which products tend to be purchased at the same time
- There are many methods (or) algorithms to perform association rule mining (or) frequent item set mining. They are
 - Apriori algorithm
 - FP-Growth algorithm

Apriori algorithm :

- Apriori algorithm uses frequent item sets to generate association rules and it is designed to work on the databases that contain transactions
- With the help of these association rule, it determines how strongly (or) how weakly two objects are connected
- Apriori algorithm was given by the R.Agrawal and Srikant in the year 1994
- It is mainly used for market basket analysis and helps to find those products that can be bought together

Steps for apriori algorithm :

- Apriori algorithm is a sequence of steps to be followed to find the most frequent item set in the given database
- A minimum support threshold is given in the problem (or) it is assumed by the user
- The steps followed in the apriori algorithm of data mining are
 - Join step
 - Prune step

Join step :

- This step generates (K+1) item set from K-item sets by joining each item with itself

Prune step :

- This step scans the count of each item in the database
- If the candidate item does not meet minimum support, then it is regarded as infrequent and thus it is removed
- This step is performed to reduce the size of the candidate item sets
- The above join and prune steps iteratively until the most frequent item sets are achieved

Ex :

- Consider the following data set and find frequent item sets and generate association rules for them
- Assume that minimum support threshold ($S = 50\%$) and minimum confident threshold ($c = 80\%$)

Transaction	List of items
T1	I1, I2, I3
T2	I2, I3, I4
T3	I4, I5
T4	I1, I2, I4
T5	I1, I2, I3, I5
T6	I1, I2, I3, I4

Solution :

Finding frequent item sets :

- Support threshold = 50%
= $0.5 * 6$
= 3
Min_support = 3

Step 1 :

- Create a table containing support count of each item present in data set – Called C1 (Candidate set)

Item	Count
I1	4
I2	5
I3	4
I4	4
I5	2

- Prune step – Compare candidate set item's support count with minimum support count. The above table shows that I5 item does not meet min_support = 3, thus it is removed and only I1, I2, I3, I4 meet min_support count
- This gives us the following item set L1

Item	Count
I1	4
I2	5
I3	4
I4	4

Step 2 :

- Join step – Generate candidate set C2 (2-item set) using L1 and find out the occurrences of 2-item set from the given dataset

Item	Count
I1, I2	4
I1, I3	3
I1, I4	2
I2, I3	4
I2, I4	3
I3, I4	2

- Prune step – Compare candidate set item's support count with minimum support count. The above table shows that item sets { I1, I4 } and { I3, I4 } does not meet min_support = 3. Thus those are removed
- This gives us the following item set L2

Item	Count
I1, I2	4
I1, I3	3
I2, I3	4
I2, I4	3

Step 3 :

- Join step – Generate candidate set C3 (3-item set) using L2 and find out the occurrences of 3-itemset from the given dataset

Item	Count
I1, I2, I3	3
I1, I2, I4	2
I1, I3, I4	1
I2, I3, I4	2

- Prune step – Compare candidate set item's support count with minimum support count. The above table shows that item set { I1, I2, I4 }, { I1, I3, I4 } and { I2, I3, I4 } does not meet min_support = 3, thus those are removed
- Only the item set { I1, I2, I3 } meet min_support count

Generate association rules :

- Thus, we have discovered all the frequent item-sets
- Now we need to generate strong association rules (satisfies the minimum confidence threshold) from frequent item sets

- For that we need to calculate confidence of each rule
- The given confidence threshold is 80%
- The all possible association rules from the frequent item set { I1, I2, I3 } are

- $\{I1, I2\} \Rightarrow \{I3\}$

$$\text{Confidence} = \frac{\text{support } \{I1, I2, I3\}}{\text{support } \{I1, I2\}} = (3/4) * 100 = 75\% \text{ (**Rejected**)}$$

- $\{I1, I3\} \Rightarrow \{I2\}$

$$\text{Confidence} = \frac{\text{support } \{I1, I2, I3\}}{\text{support } \{I1, I3\}} = (3/3) * 100 = 100\% \text{ (**Selected**)}$$

- $\{I2, I3\} \Rightarrow \{I1\}$

$$\text{Confidence} = \frac{\text{support } \{I1, I2, I3\}}{\text{support } \{I2, I3\}} = (3/4) * 100 = 75\% \text{ (**Rejected**)}$$

- $\{I1\} \Rightarrow \{I2, I3\}$

$$\text{Confidence} = \frac{\text{support } \{I1, I2, I3\}}{\text{support } \{I1\}} = (3/4) * 100 = 75\% \text{ (**Rejected**)}$$

- $\{I2\} \Rightarrow \{I1, I3\}$

$$\text{Confidence} = \frac{\text{support } \{I1, I2, I3\}}{\text{support } \{I2\}} = (3/5) * 100 = 60\% \text{ (**Rejected**)}$$

- $\{I3\} \Rightarrow \{I1, I2\}$

$$\text{Confidence} = \frac{\text{support } \{I1, I2, I3\}}{\text{support } \{I3\}} = (3/4) * 100 = 75\% \text{ (**Rejected**)}$$

- This shows that the association rule $\{I1, I3\} \Rightarrow \{I2\}$ is strong, if minimum confidence threshold is 80%

Limitations of apriori algorithm :

- Need several iterations of the data
- Uses a uniform minimum support threshold
- Difficulties to find rarely occurring events
- Some competing alternative approaches focus on partition and sampling

Challenges of frequent pattern mining :

Multiple scans of transaction database :

- The Apriori algorithm is often used for frequent pattern mining, but it needs to scan the transaction database multiple times to find frequent patterns
- Each time it scans the database, it checks whether the candidate patterns appear in the transactions
- This can be very slow, especially with large datasets

Ex :

- Imagine you have a list of customer purchases at a store
 - Transaction 1 : { Milk, Bread }
 - Transaction 2 : { Milk, Butter }
 - Transaction 3 : { Bread, Butter }
- Apriori will first scan the database to count single items like { Milk }, { Bread } and {Butter}
- Then, it will scan the database again to check combinations of items like { Milk, Bread }, { Milk, Butter } and { Bread, Butter }
- With large datasets, this can take a long time

Huge number of candidates :

- As the size of the dataset increases, the number of candidate patterns (combinations of items) increases significantly
- This can overwhelm the system with too many patterns to check

Ex :

- If you start looking for combinations of 3 items (like { Milk, Bread, Butter }), you could have a very large number of such combinations, especially when the dataset contains many items
- So, if there are 10 items, you would need to check all combinations of 3 items out of those 10, which results in 120 combinations! This is inefficient

Tedious workload of support counting for candidates :

- After finding the candidates, you need to count how often they appear in the transactions
- This support counting can be very tedious and time-consuming when there are many candidates

Ex :

- For each combination, you must check every transaction to see if it includes that combination
- If you have thousands of transactions and hundreds of candidate patterns, this becomes very slow and demanding on the system

Improving apriori efficiency :

- Apriori algorithm is a classical algorithm of association rule mining and widely used for generating frequent item sets
- This classical algorithm is inefficient due to so many scans of database
- If the database is large, it will take too much time to scan the database
- To overcome these limitations, researchers have made a lot of improvements to the apriori
- They are 4 techniques for improving efficiency of apriori algorithm are
 - Hash based technique
 - Transaction reduction

- Partitioning
- Sampling

Hash based technique :

- The hash-based technique is used to improve the efficiency of finding item sets in a dataset
- Instead of directly looking at all possible combinations of items, it uses a hash table to store and count how often each candidate item set appears

Transaction reduction :

- Imagine you're a store owner and you want to find out which products are frequently bought together by customers
- You use an algorithm called Apriori to do this
- But to make this process faster, you can use transaction reduction to narrow down the number of transactions you need to look at

Note :

- **Without transaction reduction** - You have 10,000 transactions and If you want to find items bought together (like Milk and Butter), you need to look at all 10,000 transactions, which can take a lot of time
- **With transaction reduction** - After finding that Milk and Butter are frequently bought together, you can reduce the number of transactions by skipping the ones that don't include both Milk and Butter
- Maybe now, you only need to scan 5,000 transactions instead of 10,000

Partitioning :

- Partitioning is a method used to improve the performance of finding frequent item sets (items that are often bought together) in large datasets
- Instead of scanning the entire dataset all at once, the dataset is divided into smaller, non-overlapping partitions
- Then, the algorithm processes each partition separately in two phases

Sampling :

- Sampling is a method where, instead of looking at all the transactions in your data (which might be too large), you randomly select a small sample of the data to look for frequent item sets
- This reduces the amount of data you need to process, which can save time and resources

FP-Growth algorithm : (Mining frequent item set without candidate generation)

- This algorithm is an improvement to the apriori method
- A frequent pattern is generated without the need for candidate generation

- FP-Growth algorithm represents the database in the form of a tree called a frequent pattern (or) FP tree
- This tree structure will maintain the association between the item sets
- The database is fragmented using one frequent item
- This fragmented part is called “ Pattern fragment ”
- The item sets of these fragmented patterns are analyzed
- Thus with this method, the search for frequent item sets is reduced comparatively

FP-Tree :

- Frequent pattern tree is a tree-like structure that is made with the initial item sets of the database
- The purpose of the FP tree is to mine the most frequent pattern
- Each node of the FP tree represents an item of the item set
- The root node represents null while the lower nodes represents the item sets
- The association of the nodes with the lower nodes that is the item sets with the item sets are maintained while forming the tree

Working on FP-Growth algorithm :

Step 1 :

- The first step is to scan the database to find the occurrences of the item sets in the databases
- This step is same as the first step of apriori
- The count of 1-item sets in the database is called support count (or) frequency of 1-item set

Step 2 :

- The second step is to construct the FP tree
- For this, create the root of the tree
- The root is represented by null

Step 3 :

- The next step is to scan the database again and examine the transactions
- Examine the first transaction and find out the item set in it
- The item set with the maximum count is taken at the top, the next item set with lower count and so on
- It means that the branch of the tree is constructed with transaction item sets in descending order of count

Step 4 :

- The next transaction in the database is examined
- The item sets are ordered in descending order of count

- If any item set of this transaction is already present in another branch (for example in the 1st transaction), then this transaction branch would share a common prefix to the root
- This means that the common item set is linked to the new node of another item set in this transaction

Step 5 :

- Also, count of the item set is incremented as it occurs in the transactions
- Both the common node and new node count is increased by 1 as they are created and linked according to transactions

Step 6 :

- The next step is to mine the create FP tree
- For this, the lowest node is examined first along with the links of the lowest nodes
- The lowest node represents the frequency pattern length 1
- From this, traverse the path in the FP tree
- This path (or) paths are called a conditional pattern base
- Conditional pattern base is a sub-database consisting of prefix paths in the FP tree occurring with the lowest node (suffix)

Step 7 :

- Construct a conditional FP tree, which is formed by a count of item sets in the path
- The item sets meeting the threshold support are considered in the conditional FP tree

Step 8 :

- Frequent patterns are generated from the conditional FP tree

Ex :

- Let's understand with an example how the FP growth algorithm in data mining can be used to mine frequent item sets
- Suppose we have a dataset of transactions as shown in the below

Transaction ID	Items
T1	{M, N, O, E, K, Y}
T2	{D, O, E, N, Y, K}
T3	{K, A, M, E}
T4	{M, C, U, Y, K}
T5	{C, O, K, O, E, I}

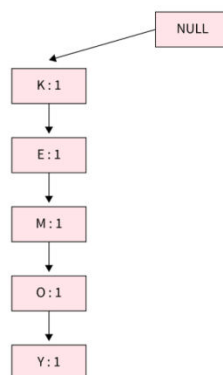
- Let's scan the above database and compute the frequency of each item as shown in the below table

Item	Frequency
A	1
C	2
D	1
E	4
I	1
K	5
M	3
N	2
O	3
U	1
Y	3

- Let's consider minimum support as 3
- After removing all the items below minimum support in the above table, we would remain with these items – { K : 5, E : 4, M : 3, O : 3, Y : 3 }
- Let's re-order the transaction database based on the items above minimum support
- In this step, in each transaction, we will remove infrequent items and re-order them in the descending order of their frequency, as shown in the below table

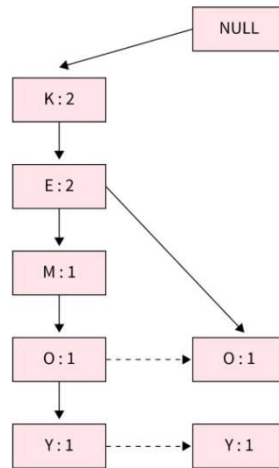
Transaction ID	Items	Ordered Item set
T1	{M, N, O, E, K, Y}	{K, E, M, O, Y}
T2	{D, O, E, N, Y, K}	{K, E, O, Y}
T3	{K, A, M, E}	{K, E, M}
T4	{M, C, U, Y, K}	{K, M, Y}
T5	{C, O, K, O, E, I}	{K, E, O}

- Now we will use the ordered item set in each transaction to build the FP tree
- Each transaction will be inserted individually to build the FP tree as shown in the below
- First transaction { K, E, M, O, Y }
 - In this transaction, all items are simply linked and their support count is initialized as 1

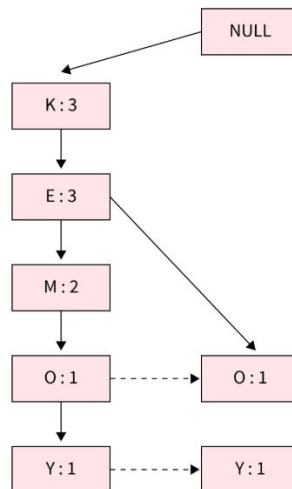


- Second transaction { K, E, O, Y }
 - In this transaction, we will increase the support count of K and E in the tree to 2

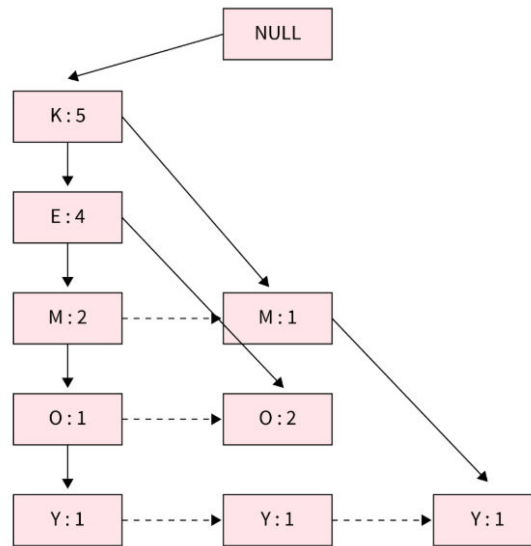
- As no direct link is available from E to O, we will insert a new path for O and Y and initialize their support count as 1



- Third transaction { K, E, M }
 - After inserting this transaction, the tree will look as shown in the below
 - We will increase the support for K and E to 3 and for M to 2



- Fourth transaction { K, M, Y } and Fifth transaction { K, E, O }
 - After inserting the last two transactions, the FP-tree will look like as shown in the below



- Now we will create a conditional pattern base for all the items
- The conditional pattern base is the path in the tree ending at the given frequent item
- For example, for item O, the paths { K, E, M } and { K, E } will result in item O
- The conditional pattern base for all items will look like as shown in the below table

Item	Conditional Pattern Base
Y	{K, E, M, O : 1}, {K, E, O : 1}, {K, M : 1}
O	{K, E, M : 1}, {K, E : 2}
M	{K, E : 2}, {K : 1}
E	{K : 4}
K	

- Now for each item, we will build a conditional frequent pattern tree
- It is computed by identifying the set of elements common in all the paths in the conditional pattern base of a given frequent item and computing its support count by summing the support counts of all the paths in the conditional pattern base
- The conditional frequent pattern tree will look like this as shown in the below table

Item	Conditional Pattern Base	Conditional FP Tree
Y	{K, E, M, O : 1}, {K, E, O : 1}, {K, M : 1}	{K : 3}
O	{K, E, M : 1}, {K, E : 2}	{K, E : 3}
M	{K, E : 2}, {K : 1}	{K : 3}
E	{K : 4}	{K : 4}
K		

- From the above conditional FP tree, we will generate the frequent item sets as shown in the below table

Item	Frequent Patterns
Y	{K, Y - 3}
O	{K, O - 3}, {E, O - 3}, {K, E, O - 3}
M	{K, M - 3}
E	{K, E - 4}

Advantages of FP-Growth :

- This algorithm needs to scan the database only twice when compared to apriori which scans the transactions for each iteration
- The pairing of items is not done in this algorithm and this makes it easier
- The database is stored in a compact version in memory
- It efficient and scalable for mining both long and short frequent patterns

Disadvantages of FP-Growth :

- FP tree is more cumbersome and difficult to build than apriori
- It may be expensive
- When the database is large, the algorithm may not fit in the shared memory

Difference between Apriori and FP-Growth algorithm :

Aspect	Apriori	FP-Growth
Pattern generation	Apriori generates pattern by pairing the items into singletons, pairs and triplets	FP-Growth generates pattern by constructing a FP tree
Candidate generation	Apriori uses candidate generation	There is no candidate generation
Process	The process is comparatively slower than FP-Growth and the runtime increases exponentially with increase in number of item sets	The process is faster as compared to apriori and the run time of process increases linearly with increase in number of item sets
Memory usage	Candidates combinations are saved in memory	Compact version of database is saved

Compact representation of frequent item set :

- Compact representation helps reduce the amount of memory and time required to store and process all the frequent item sets
- Instead of storing every possible frequent item set (which can be a large number), we store only a smaller subset that can represent all frequent item sets
- This smaller subset can then be used to generate (or) derive the larger set of frequent item sets
- One of the ways to achieve a compact representation is by using maximal frequent item sets

Maximal frequent item sets :

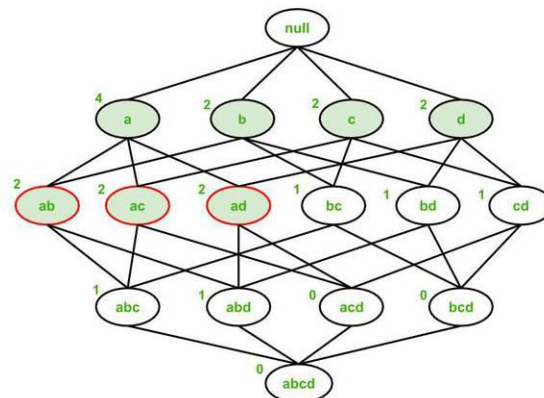
- In the Apriori algorithm, we generate frequent item sets (combinations of items that appear frequently in the transactions)
- But sometimes, the number of frequent item sets can get very large, making it hard to store and process all of them
- To solve this, we can use maximal frequent item sets
- A maximal frequent item set is simply a frequent item set that cannot be extended by adding any more items to it without it becoming infrequent (i.e., the new set won't appear often enough in the transactions)
- In simple terms, a maximal frequent item set is the largest possible combination of items that is still frequent and if you try to add any more items to it, the resulting item set will no longer be frequent

Ex:

- We have the following transactions at a grocery store as shown in the below table

TID	Items
1	a,b,d
2	a,c
3	a,b,c
4	a,d

- Let's say we are interested in frequent item sets where support threshold is 50%
- This means any item set must appear in at least 2 transactions to be considered frequent
- After analyzing the transactions, we find the following frequent item sets (those that appear in at least 2 transactions)



- a appears in transactions 1, 2, 3 and 4 → Frequent

- b appears in transactions 1, 3 → Frequent
 - c appears in transaction 2, 3 → Frequent
 - d appears in transaction 1, 4 → Frequent
- Now, let's look at pairs of items (2-item sets)
 - ab appears in transactions 1, 3 → Frequent
 - ac appears in transactions 2, 3 → Frequent
 - ad appears in transactions 1, 4 → Frequent
 - bc appears in transactions 3 → Not frequent
 - bd appears in transactions 1 → Not frequent
 - cd not appears in any transactions → Not frequent
- Now, let's look at pairs of items (3-item sets)
 - abc appears in transactions 3 → Not frequent
 - abd appears in transactions 1 → Not frequent
 - acd not appears in any transactions → Not frequent
 - bcd not appears in any transactions → Not frequent
- Finally, we look at the combination of all four items
 - abcd not appears in any transactions → Not frequent
- Now, let's identify the maximal frequent item sets
- These are the largest frequent item sets where no intermediate superset (a larger set with one more item added) is also frequent
 - ab – If we try to add any other item like “ c ”, we get the set abc, which is not frequent. So, ab is a maximal frequent item set
 - ac - If we try to add any other item like “ b ”, we get the set abc, which is not frequent. So, ac is a maximal frequent item set
 - ad - If we try to add any other item like “ c ”, we get the set abc, which is not frequent. So, ad is a maximal frequent item set
- The final maximal frequent item sets are ab, ac, ad

Note :

- Without maximal frequent item sets, we have 7 frequent item sets are { a, b, c, d, ab, ac, ad }
- With maximal frequent item sets, we have only 3 frequent item sets are { ab, ac, ad }